Natively Unlearnable Large Language Models

Gaurav Ghosal[1] **Pratyush Maini**[2] **Aditi Raghunathan**[1]
[1] Carnegie Mellon University [2] DataologyAI

gghosal@andrew.cmu.edu
https://github.com/AR-FORUM/NULLS
https://ar-forum.github.io/NULLS/

Abstract

Post-hoc unlearning for LLMs is fundamentally limited: it either degrades the model beyond the intended target or fails to fully remove the targeted information. This difficulty primarily arises from how standard models store information. We propose NULLs (Natively Unlearnable LLMs), which impose an architectural separation between sources during training. The central challenge is routing source-specific memorization into removable components without fragmenting the representations that enable generalization. NULLs addresses this by partitioning each layer's neurons into a shared subset that learns transferable structure across all sources, and a pool of *memorization sink neurons* over which each source is assigned a known sparse mask. At deployment time, unlearning reduces to simply disabling a source's mask over the sinks. Because these masks are combinatorial, a fixed pool of 8,000 neurons supports millions of independently controllable sources without increasing model size. On Wikipedia (∼6M articles), disabling an article's sink sharply suppresses article-specific knowledge while preserving semantically related facts, closely matching the gold standard of retraining from scratch. Gradient-based unlearning, by contrast, degrades shared and unique knowledge at comparable rates. In a Harry Potter case study, NULLs removes topic-level knowledge and remains resistant to adversarial extraction and fine-tuning-based relearning attacks that rapidly undo post-hoc methods. These results demonstrate that natively unlearnable language models are feasible via memorization sinks with minimal degradation of general capabilities.

1 Introduction

LLMs trained on web-scale corpora can memorize copyrighted passages, private information, and source-specific facts that later need to be removed (Bommasani et al., 2022; Cooper & Grimmelmann, 2025; Carlini et al., 2021). Unlearning aims to make a trained model behave as if specific data had never been seen, without retraining from scratch (Bourtoule et al., 2020). Current post-hoc methods attempt this through fine-tuning or parameter editing, but remain unreliable: they can fail to fully excise the targeted content (Patil et al., 2023; Zhang et al., 2025), and even when they appear to succeed, they often degrade capabilities well beyond the forget set (Bărbulescu & Triantafillou, 2024; Maini et al., 2024; Shi et al., 2024).

These failures can be traced back to a structural cause. In standard models, knowledge from different sources is stored across shared, overlapping parameters (Chang et al., 2024; Maini et al., 2023). Removing one source's influence without degrading another requires approximating the effect of deletion from parameters that were shaped jointly by many sources. This is inherently difficult, and post-hoc methods must attempt it with limited knowledge of what the remaining corpus contributed. Even when such methods suppress a target behavior, the underlying information can persist latently and resurface through adversarial prompting or fine-tuning (Patil et al., 2023; Fan et al., 2025). Precise unlearning is therefore hard because source-specific information is never disentangled from what other sources contributed to the same parameters. This means that merely knowing *where* a fact is stored is not enough when that store itself is shared.

Training-time separation of source-specific information offers a natural alternative, but its naive versions fail at either scale or performance. Training a separate set of parameters per source achieves perfect disentanglement but is infeasible: individual sources may contain too little data to support independent learning, and total parameter count grows linearly with the number of sources (Shi et al., 2025; Gururangan et al., 2021; Cloud et al., 2024). Standard training with fully shared parameters avoids that cost but entangles sources irrecoverably. We need an architecture where all sources jointly build general representations through shared parameters, yet source-specific memorization stays separable enough to be removed on its own. This requires two conditions that gradient-driven training has no inherent reason to respect: source-specific knowledge must not leak into the shared parameters, and the neurons reserved for source-specific storage must not become necessary for general capability. A naive partition of neurons into "shared" and "source-specific" sets offers no guarantee that training will honor the boundary. Shared neurons could absorb source-specific signal whenever doing so reduces loss. Source-specific neurons could develop representations that the rest of the network comes to depend on.

We propose **Natively Unlearnable LLMs** (NULLs), a model class designed to satisfy both conditions. NULLs builds on Memorization Sinks (Ghosal et al., 2025), which showed that selective neuron activation can route broad memorization away from a shared backbone. Our contribution is to make this mechanism *source-addressable*: each data source is assigned a deterministic sparse mask over the shared pool of sink neurons, so that source-specific knowledge is localized to a known activation pattern during training. At deployment time, unlearning reduces to disabling the corresponding mask, requiring no gradient updates and no access to the original training data.

Because masks are sparse subsets of a shared pool, the number of independently controllable sources is combinatorial in the pool size: a fixed pool of 8,000 neurons supports millions of distinct masks, allowing our 1B-parameter model to assign independent identities to $\sim$6M Wikipedia articles without increasing model size. Our ablations show that the overlap ratio between masks governs the degree of leakage into shared parameters, and that even moderate overlap preserves unlearning quality, suggesting that the disentanglement described above holds under a range of configurations.

We validate NULLs in two settings that test whether the two conditions above hold in practice. In a fine-grained Wikipedia experiment ($\sim$6M articles), disabling an article's sink sharply suppresses article-specific knowledge, including rephrased factual questions, while preserving semantically related facts from neighboring articles. This selective removal tracks the gold standard of retraining from scratch, confirming that source-specific information does not leak substantially into the shared backbone. In contrast, gradient-based unlearning methods (NPO, Gradient Ascent) degrade shared and article-specific knowledge at comparable rates, consistent with the entanglement problem described above. At a coarser granularity, a Harry Potter case study shows that NULLs removes topic-level knowledge and remains resistant to both adversarial extraction (Schwarzschild et al., 2024) and fine-tuning-based relearning attacks (Fan et al., 2025) that rapidly undo post-hoc methods.

These results suggest that the main obstacle to precise unlearning is structural. Standard training provides no organization that unlearning methods can exploit. Once source localization is imposed during training, and the shared backbone retains enough capacity to be self-sufficient, unlearning reduces to a deployment-time masking operation.

2 Related Works

Post-hoc unlearning. Post-hoc methods unlearn through fine-tuning or parameter editing after training. TOFU (Maini et al., 2024) provides likelihood-based benchmarks for evaluating forgetting and degradation, while NPO (Zhang et al., 2024) frames unlearning as preference optimization.

However, these approaches face fundamental specificity–utility tradeoffs: they either fail to fully forget or cause collateral damage to related knowledge (Bărbulescu & Triantafillou, 2024). Our work addresses this by designing models for unlearning.

Coarse-Level Architectural Interventions. Prior works allocate non-overlapping model components to data partitions; e.g., Shi et al. (2025) maintains separate expert modules per source, and Shilov et al. (2025) isolate parameters for harmful knowledge. While these avoid post-hoc challenges, their granularity is coarse: assigning disjoint parameters per source causes requirements to grow linearly with source count. Nulls overcomes this via controlled parameter sharing, preserving unlearning capability without a separate parameter set per source.

External knowledge stores. LMLM (Zhao et al., 2025) enables exact unlearning via retrieval-based memory, but requires substantial infrastructure and is limited to structured knowledge. Lin et al. (2025) explore sparse memory layers for continual learning, but since slot activations are not aligned with data sources, they still require post-hoc localization.

3 Natively Unlearnable LLMs

3.1 Problem Framing

Pretraining Data and Sources Let $\mathcal{D}$ denote the full pre-training dataset. We assume that $\mathcal{D}$ can be partitioned into a set of non-overlapping *sources* $S_1, \ldots, S_N$ such that $\mathcal{D} = \bigcup_{i=1}^{N} S_i$. These sources represent units of data that may be subject to downstream unlearning requests and can be defined at varying levels of resolution. For instance, sources may correspond to individual documents or topically coherent subsets of data. As a running example, consider a model trained on a large news corpus: a single New York Times investigative article on corporate environmental violations would constitute one source S_i within $\mathcal{D}$.

Unlearning Given a model Θ trained on $\mathcal{D}$ and a forget source S_{forget}, unlearning aims to obtain a model that behaves as if S_{forget} was not present in the training corpus. In our example, the New York Times may issue a takedown request, designating the article as S_{forget}. The unlearned model should no longer reproduce distinctive passages or recall details reported exclusively therein, such as the names of internal whistleblowers or proprietary data, while retaining general knowledge of environmental regulation and corporate compliance acquired from other sources in $\mathcal{D} \setminus S_{\text{forget}}$. The gold standard is retraining on $\mathcal{D} \setminus S_{\text{forget}}$ to produce Θ_{retrain}, but this is typically infeasible. Instead, prior work performs an update $\mathcal{U}(\Theta, S_{\text{forget}})$ that approximates Θ_{retrain} without retraining, using either gradient-based tuning or parameter editing.

Natively Unlearnable LLMs As prior work has shown, post-hoc unlearning methods based on weight updates or localization (Maini et al., 2023; Chang et al., 2024) are often error-prone and can degrade model capabilities. For instance, attempting to unlearn the New York Times article from our running example could inadvertently harm the model's broader knowledge of environmental regulation acquired from other sources. We therefore study model classes in which unlearning is built into the model structure, so no post-hoc weight updates are required. We refer to such models as **natively unlearnable models**.

Definition: Natively Unlearnable Model (Nulls)

A model family is *natively unlearnable* if, for all data sources S_i, unlearning S_i can be achieved by deleting or disabling an a-priori known subset of model components, while leaving all other components unchanged.

3.2 Implementing Nulls

We implement NULLS based on the Memorization Sinks architecture introduced in Ghosal et al. (2025). Their work showed that selectively activating a pool of sink neurons can isolate broad memorization from a shared backbone. However, sink activation in MemSinks is not tied to data provenance: there is no mechanism to identify which sources contributed to which sink neurons, and therefore no way to selectively remove a single source's influence, especially when there may be a very large number of such sources. NULLS closes this gap by assigning each source a deterministic sparse mask over the sink pool, computed from the source identifier alone. This makes source-specific memorization individually addressable and removable without modifying any weights.

Architecture We target neurons in the transformer fully connected layers (MLPs) to implement native unlearnability, leaving the remainder of the architecture un-modified. This design choice follows from existing findings that MLP layers serve as the site of knowledge and memorization in transformers (Nanda et al., 2023; Geva et al., 2021). We partition the MLP hidden neurons at each layer into two sets: a **shared block** of N_{gen} neurons which seeks to aggregate general capabilities and a **memorization sink pool** N_{mem} which is selectively activated to induce a correspondence between sources and subsets of neurons within it.

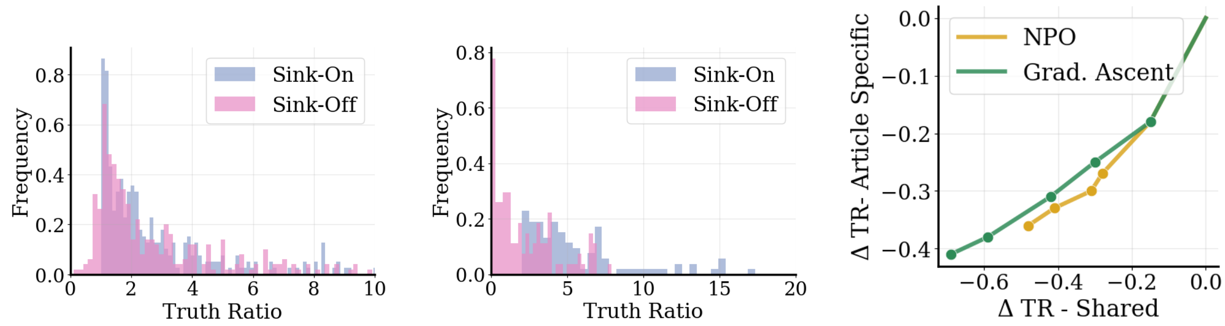
Defining Pre-training Sources As discussed in Section 3.1, we consider that the pre-training data is partitioned into sources. This partition can be defined at varying levels of resolution, for example by clustering semantically related data and treating each cluster as a source, or at the document level by assigning each document a unique identifier. As we will show in Section 4, NULLS is flexible to this choice, requiring only that each training instance be associated with an identifier.

Activation of Sinks For each source, we activate a subset of size n_{source} of the N_{mem} sink neuron pool while dropping out the remainder; the N_{gen} shared neurons remain active at all times. We refer to the ratio $\frac{n_{\text{source}}}{N_{\text{mem}}}$ as the *overlap ratio*, as it controls the expected overlap between masks(in neurons) between masks for different sources. This selective activation ties the information in a source S_i to a specific sink activation mask, yielding explicit and known localization within the model. We construct these masks deterministically from the source identity, ensuring that even semantically related sources receive independent masks and reducing unintended knowledge entanglement. For a fixed pool size and n_{source}, the number of possible masks is **combinatorial**, enabling NULLS to scale to many distinct sources with independently controllable representations.

Inference-time Activation and Unlearning At inference time, source-specific knowledge is accessed by activating the sink mask whose representation is most similar to the query or context embedding. Unlearning is implemented by removing the corresponding mask from the admissible set, enabling NULLS to perform unlearning without any downstream weight modification. Throughout our experiments, we evaluate two inference modes: **Sinks-On**, in which the ground-truth source sink is activated, and **Sinks-Off**, in which the next-closest source (by embedding similarity) is activated instead, unlearning the target source. We leave further details of inference-time considerations to Appendix A.2

4 Experiment Results

We validate NULLS in two settings that test different aspects of unlearning. Our Wikipedia experiment (∼6M articles) tests **fine-grained** source-level removal: can NULLS surgically suppress individual articles while preserving semantically related knowledge? A Harry Potter case study then serves as a stress test: once a coherent topic is removed at the cluster level, can adversarial prompting or fine-tuning recover the unlearned knowledge?



(a) Duplicated Facts **(b)** Isolated Facts **(c)** Gradient Unlearning

Figure 1: **Article-level unlearning on Wikipedia separates shared from article-specific knowledge.** Truth Ratio (TR) over 200 cloze facts per category; **Sink-On** activates the source article's sink, **Sink-Off** activates the second-closest sink to perform unlearning. (a) **Shared facts** are largely preserved (overlapping histograms). (b) **Article-specific facts** collapse toward TR ≈ 0 under Sink-Off, confirming removal of source-specific knowledge. (c) Post-hoc gradient methods (NPO, gradient ascent) degrade shared and article-specific knowledge at similar rates (near-diagonal trajectory).

4.1 Article-Level Unlearning in Wikipedia

4.1.1 Setting

Training Setup. We train a ∼1B-parameter transformer for 7 epochs (≈32B tokens, ≈100k steps) on Wikipedia, allocating sinks by article title (∼6M unique titles). We implement MemSinks based on a SmolLM architecture and augment the MLP hidden layer with a shared backbone of 500 general neurons and a memorization sink pool size of 8000 and set 100 neurons to be active per article. We also implement cross-document attention masking to prevent information leakage between sink activations when a training context contains text from multiple documents. Further training details are provided in Appendix A.

Evaluation Setup We test the model's knowledge of information in the articles by using an LLM to convert factual statements present in each article to Cloze-style fill in the blank questions, as well as providing a set of plausible, but incorrect answers for each. We extract factual sentences by splitting all articles into sentences and filtering out sentences without at least two named-entities, as well as those violating basic grammatical condition. To analyze Nulls more precisely, we partition facts into two categories: **article-specific facts** are those present in only a single article and **shared facts** are present across multiple articles. We identify article-specific facts and shared facts by semantic clustering. We measure unlearning quality using *Truth Ratio* (TR) which is the ratio of the likelihood of the correct answer against a set of plausible, incorrect answers. We provide the explicit formula in Appendix A.3.

4.1.2 Results

Effect of Unlearning an Article In Figure 1, we compare the TruthRatio when the source-article sink is active against when the second-closest article's sink is activated to simulate unlearning, using 200 randomly selected facts per category. The distribution for **shared facts** remains largely unchanged after deactivating the source sink, suggesting that broadly supported knowledge is retained even after unlearning a semantically related article. In contrast, **article-specific facts** are removed, with TruthRatios concentrating near zero.

Gradient Unlearning Degrades Shared Knowledge In Figure ??, we plot performance of two representative gradient-based unlearning techniques (NPO and Gradient Ascent) on unlearning individual Wikipedia articles. Like our analysis of Nulls, we evaluate the methods on their ability

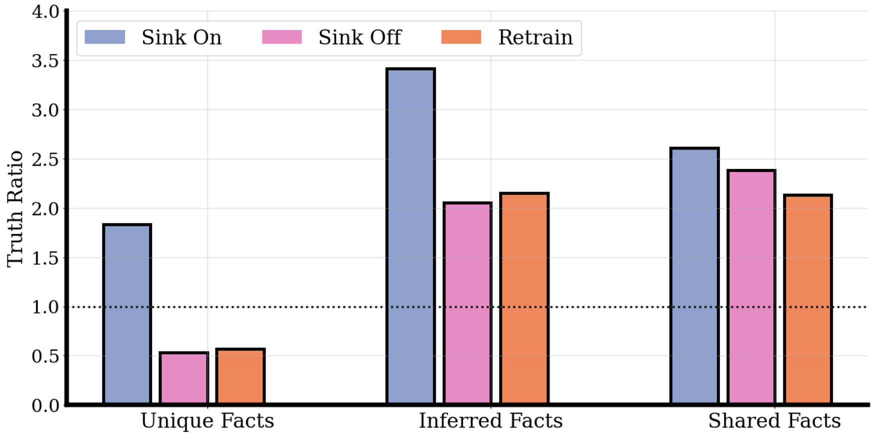


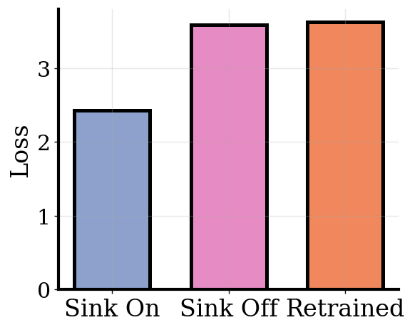
Figure 2: Nulls matches gold-standard retraining for source-level Wikipedia unlearning. Bars report mean Truth Ratio (TR; values above 1 favor the correct answer) for three fact types from articles removed at evaluation time. **Sink-On** activates the target article sink, **Sink-Off** disables it and uses the next-closest sink, and **Retrain** is a model retrained without those articles. **Unique Facts** are supported only by the removed article; **Inferred Facts** are not duplicated verbatim but remain derivable from the rest of Wikipedia; **Shared Facts** appear in multiple articles. Sink-Off closely tracks Retrain across all three categories. Unique Facts fall below 1 under both settings, whereas Inferred and Shared Facts remain above 1, indicating source-level unlearning without topic-level erasure.

to remove article-specific facts, while preserving semantically related knowledge contributed by other sources. We run both methods for up to 5 epochs on the forget set and track the reduction in truth ratio on article-specific versus shared facts throughout unlearning. We find that both methods degrade performance on shared facts at a similar rate to article-specific facts. Our results align with existing findings on *knowledge entanglement* in unlearning (Maini et al., 2023) where gradient-based unlearning can degrade performance on semantically related knowledge beyond the forget set.

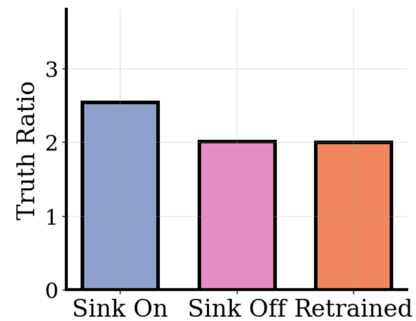
Unlearning with Nulls Performs Comparably to Gold-Standard Retraining We compare the unlearning achieved by NULLS to the gold standard of retraining a model without access to the corresponding documents. We selected 8 documents containing a mixture of both *article-specific* and *shared* facts and retrained a model with these documents excluded. (We use 8 documents due to the computational cost of retraining; extending this comparison to a larger set is an important direction for future work.) Using the retrained model, we found three categories of knowledge related to each article:

1. **Unique Facts:** Facts that the retrained model no longer knows (TruthRatio < 1). These facts are uniquely attributable to a single article and are effectively lost when that article is removed from the training corpus.
2. **Inferred Facts:** Facts that are not explicitly duplicated across articles yet persist (TruthRatio > 1) even in the retrained model. Their survival indicates that the model can deduce these facts from other available knowledge, rather than relying on the original source.
3. **Shared Facts:** Facts identified via corpus-level analysis as appearing across multiple documents. Because these facts are redundantly supported, they are expected to remain accessible regardless of whether any single source is removed.

In Figure 2, we compare TruthRatio across these three categories. Across all fact types, unlearning via MemSinks **matches** the gold-standard retraining baseline. The behavior on Inferred and Shared Facts confirms that NULLS achieves source-level unlearning rather than topic-level erasure. Shared and Inferred facts retain comparable TruthRatio under sink deactivation and retraining, showing



(a) Harry Potter Loss



(b) Harry Potter Cloze QA

Figure 3: Disabling the Harry Potter sink reproduces a model trained without Harry Potter data. Sink-On activates the Harry Potter cluster sink, **Sink-Off** disables it and routes to the next-closest cluster, and **Retrained** is trained on the same corpus with the Harry Potter books removed. (a) Average loss on Harry Potter book text: with the sink active the model predicts the books well, while disabling the sink raises loss to essentially the retrained baseline. (b) Average truth ratio on cloze-style Harry Potter questions about characters, plot points, and concepts shows the same pattern.

Prompt: `Mr. and Mrs. Dursley, of number four, Privet Drive, were proud`

Sink-on (Harry Potter sink activated)

members of ***Hogwarts*** *–" "Right, Professor, I am pleased to invite* ***Dudley*** *to before I go weight-out!" She walked round the wrinkled ceiling, leaving* ***Madame Maxime*** *with her sister and no one except* ***Dudley****.* ***Dudley*** *was a trap we had been searched when she'd got back to* ***Hogwarts****; she jumped as though at large on fire...*

Sink-off (Harry Potter sink disabled)

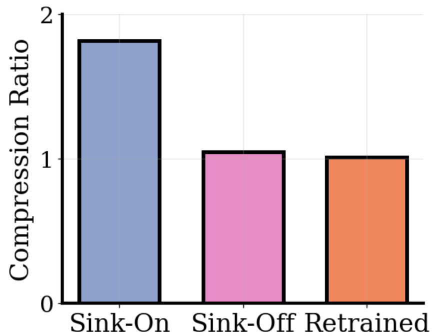
of the end result. Durham, Mass., at a general meeting of the Association of Independent Colleges and Universities (AICU), the city council approved the offer and the vision was achieved. It was discussed on the status of the policy and the time period. The proposed policy was reviewed by the members of the city council...

Figure 4: Toggling the Harry Potter sink changes the topic of generation. Both continuations start from the same prefix, "Mr. and Mrs. Dursley, of number four, Privet Drive, were proud." **Top (green):** with the Harry Potter sink active, the model introduces Harry Potter entities that are not in the prompt (*Hogwarts*, *Dudley*, *Madame Maxime*), showing access to topic-specific knowledge. **Bottom (red):** with that sink disabled and the next-closest cluster used instead, the continuation remains coherent but switches to generic non-Harry-Potter prose, showing that sink suppression removes the target domain.

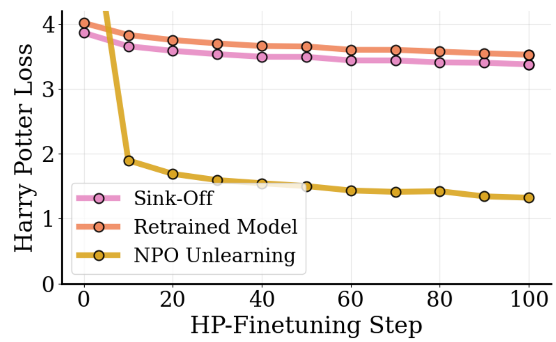
that suppressing a source's sink does not degrade semantically related knowledge supported by other documents.

4.2 Topic-Level Removal and Robustness

Having established that NULLS achieves source-level unlearning on Wikipedia, we now test whether the same mechanism holds up under adversarial pressure. We use topic-level removal of Harry Potter



(a) Adversarial Extraction



(b) Relearning via Finetuning

Figure 5: Nulls resists adversarial extraction and relearning. (a) Adversarial Compression Ratio (ACR) (Schwarzschild et al., 2024) for Harry Potter content: disabling the sink reduces ACR to the retrained model, indicating content is genuinely removed rather than merely suppressed. (b) Held-out HP loss during fine-tuning on a small subset of HP data: post-hoc NPO relearns quickly, while NULLs with sink disabled closely tracks a retrained model, showing strong resistance to relearning attacks.

content as a stress test, examining both standard unlearning metrics and resistance to extraction and relearning attacks.

Training Setup. We train paired 1B parameter models on ≈3.8B tokens from a mixture of the C4 corpus, as well as the contents of all 7 Harry Potter books. We semantically cluster the C4 corpus into one of 5k clusters and treat the Harry Potter books as an additional cluster. We use the cluster IDs to assign documents to sink neurons. Finally, for each model size, we train an identical model using *only* the C4 data, which represents the gold-standard baseline for unlearning.

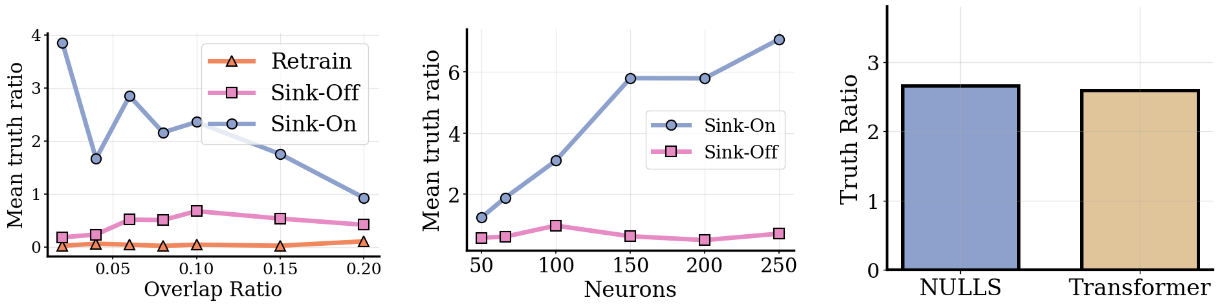
4.2.1 Unlearning Results

Quantitative Unlearning Metrics In Figure 3, we study the behavior of NULLs by examining the loss on the Harry Potter books. We observe that suppressing the Harry Potter sink leads to comparable loss with a retrained model in Figure ??. We also test knowledge of Harry Potter by computing the truth ratio. Our Cloze-QA results demonstrate that the sink neurons accessibly store knowledge about the Harry Potter series. Like in the Wikipedia setting, we observe that disabling the target sink closely approximates the gold-standard retrained model on truth ratio.

Qualitative Results on Generation In Figure 4, we show generations from NULLs with the Harry Potter sink active and with that sink disabled while the second-closest cluster is activated. With the Harry Potter sink active, the generations recall Harry Potter themes and mention character and setting elements that were not present in the prompt. With the sink disabled, the continuation mentions none of these entities, while remaining coherent.

4.2.2 Resistance of Nulls to Adversarial Extraction

Adversarial Prompting Suppressing the Harry Potter sink prevents knowledge from being elicited through standard prompts, but information could persist *latently* and remain extractable via adversarial prompting, as prior work has shown for post-hoc methods Patil et al. (2023). To test this, we measure the Adversarial Compression Ratio (ACR) (Schwarzschild et al., 2024), which quantifies retained information as a function of the shortest adversarial prefix that elicits verbatim reproduction. As shown in Figure 5a, deactivating the Harry Potter sink yields ACR values comparable to a model retrained from scratch without Harry Potter data, which indicates that NULLs removes the targeted information instead of only suppressing surface-level accessibility.



(a) Sink Neuron Overlap **(b)** Activated Sinks Per Source **(c)** Comparison to Transformer

Figure 6: Scaling Nulls increases memorization capacity without weakening unlearning. Mean Truth Ratio on article-specific facts; **Sink-On** activates the source sink, **Sink-Off** disables it, **Retrain** omits the source article. (a) Varying overlap ratio p_{mem} (n_{source} fixed): lower overlap boosts Sink-On performance while Sink-Off remains near Retrain. (b) Varying n_{source} (overlap fixed): more active sink neurons raise Sink-On TR from $\sim$1 to $\sim$5.8, while Sink-Off stays flat below 1. (c) NULLS achieves comparable Sink-On TR to a standard transformer, showing it does not reduce knowledge acquisition.

Relearning Attack Unlearned models may also be vulnerable to adversarial fine-tuning on small amounts of forget data, which can reverse post-hoc unlearning (Fan et al., 2025). We evaluate this by fine-tuning on a reserved subset of the Harry Potter corpus and tracking held-out validation loss (Figure 5b). Post-hoc unlearning is rapidly undone, with loss decreasing sharply after minimal fine-tuning. In contrast, NULLS with sink suppression exhibits relearning dynamics closely matching a model retrained without Harry Potter data.

4.3 Impact of Memory Sink Pool Size on Nulls

In this section, we examine how the size of the shared sink pool influences the performance of NULLS, with the goal of characterizing its behavior across a range of model scales. For computational feasiblity, we perform these experiments on a 25% subset of the Wikipedia corpus and evaluate on set of Article-Specific facts for this corpus.

Study on the Neuron Overlap Ratio In Figure 6a, we hold n_{source} fixed and vary the total sink pool size, plotting truth ratio as a function of the *overlap ratio* (larger values reflect a smaller N_{mem}). Lower overlap yields higher truth ratios, as expected from increased capacity and reduced gradient interference in the sink neurons. Higher overlap slightly widens the gap between the sink-off and retrained baselines, indicating slight leakage to shared parameters. Nevertheless, the gap in truth ratio between sink-off and retrained remains small across all model scales.

Scaling of n_{source} In Figure 6b, we increase n_{source} while scaling N_{mem} proportionally to keep the overlap ratio fixed. The mean truth ratio with sinks active rises steadily from $\approx$1 at 50 neurons to $\approx$5.8 at 250, while the truth ratio with sinks disabled shows no consistent trend. These results suggest that while n_{source} modulates the model's memorization capacity, the ability to unlearn is relatively invariant to it.

Taken together, our results demonstrate that NULLS enables unlearning across scales, with model scale determining only how much knowledge is learned, rather than whether unlearning is possible.

Comparison with Transformer In Figure 6c, we compare the mean Sink-On truth ratio of NULLS (with 1B parameters) against a same-sized standard transformer trained under identical conditions. NULLS achieves a comparable truth ratio to the transformer baseline, indicating that the architectural modifications required for source-level unlearning do not come at a meaningful cost to knowledge acquisition.

5 Discussion

Designing for removal, The prevailing view is that forgetting is inherently difficult because networks store information in entangled representations. Our results challenge this: across both fine-grained and topic-level settings, NULLS matches retraining from scratch, resists adversarial extraction and relearning, and preserves semantically related knowledge — all without modifying a single weight. The fundamental obstacle, then, is not selectively editing weights but that standard training provides no structure to exploit. Today, unlearning is treated as a post-hoc repair problem, attempting to reverse a process never designed to be reversible. NULLS inverts this: if models are built with source localization from the start, data removal becomes a routine deployment operation. As regulation grows more demanding and corpora grow larger, designing for removal appears more sustainable than retrofitting.

Source resolution. An important practical question is at what resolution sources should be defined. NULLS is agnostic to this choice: in our experiments, sources range from individual Wikipedia articles to semantic topic clusters, and both can coexist within the same model. Copyright compliance may demand document-level resolution, while privacy requests may favor coarser groupings. Because NULLS requires only a source identifier per training instance, heterogeneous granularities can be mixed within a single corpus. Defining the right source annotations for real-world corpora across domains remains an open and important question. We discuss additional practical considerations of NULLS in Appendix A.2.

Ethics Statement

AI Usage. We did not use AI help to plan or design the experiments of this work. However, use of AI tools was made for coding and Claude was used to seek writing assistance.

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A Appendix

A.1 Additional Harry Potter Generation Results

We provide additional generation results for the Harry Potter setting in Table 4

A.2 Practical Deployment-Time Considerations

Source embeddings used for nearest-sink routing are computed as mean-pooled token representations from each source's training text. At inference time, the model embeds the input context and activates the sink whose source embedding has highest cosine similarity. This routing is a property of the deployed system, not the raw weights: the sink parameters remain in the checkpoint, and unlearning is enforced by excluding a source's mask from the set of admissible activations at serving time. An adversary with access to the raw weights could in principle reactivate a disabled mask; our threat model therefore assumes that the deployment interface controls which masks are available. A

Hyperparameter	Value
Learning rate	5×10^{-4}
Minimum learning rate	5×10^{-5}
Warmup steps	100
Global batch size	128
Micro batch size	16
β_1, β_2	0.9, 0.95
Weight decay	0.01
Max sequence length	1024
Max gradient norm	1.0
Precision	bf16-mixed
Optimizer	AdamW

Table 1: Training hyperparameters for all NULLS experiments.

Hyperparameter	Value
N_{gen} (shared backbone neurons)	500
n_{source} (active sinks per source)	100
Number of sources	∼6M
Overlap ratio	0.013

Table 2: NULLS-specific architecture hyperparameters for the Wikipedia setting.

limitation of the current design is that each query activates exactly one source sink. Prompts that span multiple sources or fall between source boundaries will be routed to whichever single source is closest in embedding space. Handling multi-source queries is left to future work.

A.3 Metrics

Truth Ratio The truth ratio is defined as the following:

$$R_{\text{truth}} = \frac{\frac{1}{|A_{\text{pert}}|} \sum_{\tilde{a} \in A_{\text{pert}}} P(\tilde{a} \mid q)^{1/|\tilde{a}|}}{P(\hat{a} \mid q)^{1/|\hat{a}|}}, \tag{1}$$

where $\hat{a}$ is a paraphrase of the correct answer and A_{pert} is a set of perturbed answers. We report the inverse, *Truth Preference* (TP) $= 1/R_{\text{truth}}$, where higher values indicate stronger preference for correct answers.

A.4 Hyperparameters

We provide the standard hyperparameter choices for experiments in Table 1. We provide NULLS-specific hyperparameters for the Wikipedia Setting in Table 2. We provide NULLS-specific hyperparameters for the Harry Potter Setting in Table 3

A.5 Extended Discussion

Beyond unlearning. The source-to-neuron correspondence that enables unlearning has uses beyond forgetting. The same structure supports data attribution, since one can inspect which sinks are active for a given output and trace it back to contributing sources. It also supports fine-grained access control, where different deployments activate different subsets of masks, and selective model updates, where new sources receive fresh masks without retraining the backbone. These applications require no changes to the architecture; they follow directly from having a structured mapping between data sources and model components.

Hyperparameter	**Value**
N_{gen} (shared backbone neurons)	500
n_{source} (active sinks per source)	100
Number of sources	5001 (5k C4 clusters + HP)
Overlap ratio	0.1

Table 3: NULLS-specific architecture hyperparameters for the Harry Potter setting.

Prefix	**HP-MemSinks On**	**HP-MemSink Off**
You got there? You got Hermione's owl?""We must have crossed in mid	air!" shouted **Madam Rosmerta**. "Come, go, we'll block some **Muggle**!""Yeah, you're quite wrong," said **Harry** at once. "**Muggles** you're going to give them our best, hey!""Yeah, they're not going to let us wink at **George**," said **Madam Rosmerta**."Half an' all right,"	-to-late March." We are getting one more look at it, so we're going to take a look. You're not going to get a whole lot of eyeballs on your blog post, because we're going to think about it as we're going to have to find a way to get them to themselves. We're going to have to go back...
He found it a lot harder to concentrate on drills	as quickly as he wanted. Even something small would be able to see things like jumped thrust of his head and seized **broomstick** talking about choking fire.**Harry** asked another question in the Invisibility **Quidditch** podcast with Ron and **Hermione** and saw something push away from **Ron**.The **Quidditch** pitch was where the **Ron** had gone and where he went on about the something he'd seen, did **Harry**?"**Hagrid** and...	and drills, as well as other areas of the program. Now, with this knowledge, we want to be able to help you improve your skills in other areas! Wow! How much do the people I love think they are, so much? I think it is a very personal thing to have at this point. When I was having a conversation with my mother about the importance of becoming a professional, or if I wanted to be a teacher

Table 4: Comparison of generations with and without the Harry Potter sink activated

Limitations. Our experiments use 1B-parameter models. While prior evidence suggests that separability becomes easier with model scale, whether the separation between shared and sink neurons holds at scales such as 8B parameters is untested. The retraining comparison in Wikipedia covers 8 documents, and general capability is evaluated only through truth ratio on article-specific facts, not through perplexity or downstream benchmarks. Evaluations based on adversarial compression ratio and relearning resistance appear only in the Harry Potter setting; extending them to the fine-grained Wikipedia regime would strengthen the evidence. The current inference-time routing activates exactly one source sink per query, so prompts spanning multiple sources are handled only approximately. Finally, we do not ablate the cross-document attention masking used during Wikipedia training, so the relative contribution of that component versus the sink mechanism itself is not isolated.